

Curated Research Articles

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- **Ferroelectrically Modulated Composite Polymer Electrolyte Enabling Accelerated Ion Transport and Stable Interfaces for Solid-State Lithium Metal Batteries** — score: 1.000 The article presents a novel ferroelectrically modulated composite polymer electrolyte designed to enhance ion transport and stabilize interfaces in solid-state lithium metal batteries. By integrating ferroelectric materials, the study demonstrates improved lithium-ion conductivity while mitigating issues such as anion migration and interfacial instability, resulting in solid-state batteries that can cycle over 300 times with enhanced electrochemical stability and safety. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Unlocking Mg²⁺ Transport in Magnesium Batteries: Aromatic Architectures and Systemic Co-Design** — score: 1.000 The article explores the innovative use of aromatic and conjugated materials to enhance Mg²⁺ transport in magnesium batteries, addressing key challenges like interfacial instability and ion desolvation. It highlights the role of PDI-based cathodes and carefully designed interphases in improving performance, advocating for a systematic co-design approach that integrates cathode, anode, and electrolyte chemistry to optimize battery efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Decomposition of Co-Intercalated Solvents as the Overlooked Driver of Graphite Cathode Failure in Dual-Ion Batteries** — score: 1.000 This study identifies the decomposition of co-intercalated solvents as a major cause of graphite cathode failure in dual-ion batteries, challenging existing failure models that focus on volume changes during ion intercalation. The research highlights the significance of solvent exclusion in designing more stable graphite cathodes, revealing that controlling solvent intercalation can enhance battery performance and longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Spatiochemical Segregation in Porous Lithium–Metal Interphases** — score: 1.000 This article explores spatiochemical segregation in lithium-metal interphases within porous structures, examining how spatial distribution impacts electrochemical performance. The findings contribute to understanding interphase stability and efficiency in lithium-based batteries, highlighting the significance of material design in optimizing battery functionality. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Regulating Zn²⁺ Transport via Potassium-Intercalated V₂C MXene and Electrolyte Engineering for Advanced Aqueous Zinc-Ion Batteries** — score: 1.000 This article discusses the development of advanced aqueous zinc-ion batteries by improving Zn²⁺ transport through the use of potassium-intercalated V₂C MXene and optimizing electrolyte conditions. The proposed strategies aim to enhance battery performance and stability, showcasing innovative approaches in energy storage technology. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Prelithiated SiO_x/Graphite-NMC811 Cells: Capacity Loss, Impedance Rise and Hidden Degradation Pathways Revealed Using Three-Electrode Diagnostics** — score: 1.000 The study investigates the performance degradation of prelithiated SiO_x/Graphite-NMC811 battery cells by employing three-electrode diagnostics, which reveal insights into capacity loss, impedance increases, and previously unrecognized degradation mechanisms affecting their efficiency. The findings highlight the complexities of battery behavior under operational conditions and could inform future improvements in lithium-ion technology. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Thermal Fusion of Lithium-Philic Fluorinated Bi₂O₂CO₃ to Derive Heterogeneous SEI and Dendrite-Free Li Plating for the First Ah-Level Fluoride Conversion Pouch Cells** — score: 0.900 This study introduces a novel method using thermal fusion of fluorine-doped Bi₂O₂CO₃ with lithium metal to create a lithium-philic artificial solid electrolyte interface (ASEI), which enhances lithium-ion deposition while preventing dendrite formation. The resulting heterostructure improves electrochemical stability and has led to the development of the first Ah-level fluoride conversion pouch

cells, demonstrating impressive cycling performance and capacity retention. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Mechanically Adaptive Ultrathin Interphase Inspired by Sliding Filament for Highly Reversible Zinc Metal Anodes** — score: 0.900 The article presents a novel ultrathin interphase inspired by biological muscle sliding filaments, constructed from graphene oxide nanosheets interconnected by dynamic hydrogen bonds. This design achieves a balance between high mechanical rigidity and flexibility, effectively suppressing dendrite growth and accommodating volume changes in zinc metal anodes. As a result, the modified anodes show exceptional electrochemical performance, with over 10,000 cycles and a coulombic efficiency nearing 100%, highlighting the effectiveness of the adaptive interphase in enhancing battery stability and longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Biomimetic Zwitterionic Hydrogel Electrolyte Enables Highly Reversible Zn Anodes** — score: 0.800 This study introduces a biomimetic zwitterionic hydrogel electrolyte, PAMC, for Zn-based batteries, enhancing their performance by optimizing Zn²⁺ solvation and minimizing dendrite formation and side reactions. The PAMC electrolyte achieves a high ionic conductivity and impressive cycling stability in Zn cells, making it suitable for integration with solar cells, thereby advancing the potential for sustainable energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Insight and regulation of interfacial coordination chemistry of high-voltage LiCoPO₄ cathode via functionalized carbon layer anchoring for robust surface passivation** — score: 0.800 The article explores the enhancement of lithium cobalt phosphate (LiCoPO₄) cathodes by addressing their shortcomings in Coulombic efficiency and capacity through the implementation of a functionalized carbon layer. This innovative approach aims to improve interfacial coordination chemistry and provide robust surface passivation, potentially boosting the performance of high-voltage battery systems. Journal: *RSC - EES Batteries latest articles*
- **All-MXene Co-Design Enables Fast Iodine Redox and Ultrastable Zinc Anode in Aqueous Zn-I₂ Batteries** — score: 0.800 This article presents a novel all-MXene approach for enhancing the performance of zinc-iodine batteries by utilizing fluorine-rich MXene materials to simultaneously improve iodine redox kinetics and inhibit dendrite formation in zinc anodes. The developed method achieves remarkable electrochemical stability, with the battery retaining 72.8% capacity over 60,000 cycles, while also maintaining high capacity and efficiency, thus significantly advancing the reliability and performance of Zn-I₂ systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Open-Structured Zinc Hydroxide Sulfate Formation via Integrated Cathode Design Enabling Highly Reversible Ah-Level Zn-MnO₂ Batteries** — score: 0.800 The study introduces a novel Bi₂O₃ regulation strategy to form open-structured zinc hydroxide sulfate (ZHS) in Ah-level zinc-manganese dioxide batteries, aimed at addressing ZHS deposition challenges in large-scale applications. By utilizing a Bi₂O₃-modified cathode synthesized through high-energy ball milling, the method promotes sparse ZHS growth, enhancing charge transfer and battery performance, achieving a reversible capacity of 338.2 mAh g⁻¹ and excellent stability over extended cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dual-Doping Induced Synergistic Functionality in a Perovskite Air Electrode for Balanced Performance and Durability in Reversible Protonic Ceramic Cells** — score: 0.800 The article presents a dual-doping strategy that incorporates Ti⁴⁺ and Nb⁵⁺ into the perovskite structure of PrBaCo₂O₅₊, leading to a highly effective air electrode, PrBaCo_{1.8}Ti_{0.1}Nb_{0.1}O₅₊. This novel composition significantly improves electrocatalytic activity, proton conductivity, and stability under challenging operational conditions, achieving impressive performance metrics in both fuel cell and electrolysis modes, while also maintaining stability for over 380 hours. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Decoupled yet Synergistic Microdomains via In Situ Phase Separation Enabling Ultra-Stable Zinc Metal Anodes** — score: 0.800 The study presents a novel polymeric interphase for

zinc metal anodes created through in situ phase separation, combining tough polymer-rich domains with elastic, zinc-ion conductive regions. This design significantly enhances mechanical stability and ion transport, resulting in highly reversible zinc cycling with over 2400 hours of lifespan and exceptional durability at high discharge depths, marking a major advancement in zinc anode technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Regulating Electronic Microenvironments in Fully-Conjugated Covalent Organic Framework for High-Rate Ion Storage** — score: 0.800 The article presents a novel fully-conjugated covalent organic framework (BTTP-COF) designed to enhance lithium-ion battery performance by regulating its electronic microenvironment through structural evolution. This innovative approach improves electron transport and decreases ion diffusion resistance, resulting in higher active site availability and a specific capacity of 297 mAh g⁻¹ with outstanding cycling stability, making it a promising candidate for high-rate energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Monitoring of electro-chemo-mechanics in Li metal electrodes by optical fibre sensors** — score: 0.800 The article discusses the development of an optical fiber sensor capable of monitoring the electro-chemo-mechanical behaviors of lithium metal electrodes. This innovation aims to enhance the understanding and performance of lithium batteries, potentially leading to improvements in energy storage technology. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Crystal facet engineering empowers zinc materials with diverse facet orientations: preparation strategies, performance advantages, and application challenges** — score: 0.800 This review article explores the relationship between the structure and performance of zinc facets, detailing various fabrication techniques for creating dendrite-free zinc anodes. It highlights the advantages of different facet orientations in enhancing electrochemical performance and addresses the challenges in optimizing these materials for practical applications. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Local Electron Spin-State Engineering at Fe Sites for Highly Reversible Sodium-Ion Batteries** — score: 0.800 The study presents a novel carbon-coated sodium iron sulfide composite (Na₆FeS₄/C) that employs spin-state engineering to enhance the performance of sodium-ion batteries. This approach improves structural stability and accelerates reaction kinetics, resulting in high reversible capacity, excellent rate performance, and impressive cycling stability, demonstrating the potential of engineered Fe sites for effective sodium energy storage. Journal: *Wiley: Small Methods: Table of Contents*
- **[ASAP] Framework Electronegativity Governs Interfacial Transport Kinetics in Lithium-Metal Batteries** — score: 0.800 The article discusses how the framework electronegativity of materials influences the kinetics of interfacial transport in lithium-metal batteries. The findings suggest that understanding this relationship can enhance battery performance, potentially leading to more efficient energy storage solutions. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Identifying Electrolyte Reduction Intermediates in Lithium Metal Batteries with Spin Trapping** — score: 0.800 The article discusses the use of spin trapping techniques to investigate and identify reduction intermediates of electrolytes in lithium metal batteries. This research aims to enhance the understanding of decomposition processes in battery systems, which is crucial for improving their performance and safety. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Revealing the Structural Origins of Ionic Conductivity Degradation in Li₆PS₅Cl: A Detailed Spectroscopic Study of Moisture-Induced Chemical and Crystallographic Transformations** — score: 0.800 This study investigates the impact of moisture on the ionic conductivity of Li₆PS₅Cl by examining its chemical and crystallographic changes through detailed spectroscopy. The findings reveal how exposure to moisture leads to structural degradations that significantly impair the material’s performance in energy applications. Journal: *ACS Applied Energy*

- **[ASAP] In Situ Nanoscale Imaging of Cultivated, As-Grown Diatom Frustules Electrochemical Lithiation/Delithiation in Li Ion Battery Anodes** — score: 0.800 The article discusses the use of in situ nanoscale imaging techniques to study the electrochemical behavior of diatom frustules as anodes in lithium-ion batteries. It highlights how these naturally occurring structures, when cultivated and grown, exhibit unique properties that influence their lithiation and delithiation processes, potentially leading to advancements in battery technology. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Enhanced Electrochemical CO₂ Capture Through Galvanostatic Protocol Optimization of MnO₂-Based Proton Intercalation Systems** — score: 0.700 The study demonstrates that optimizing electrochemical cycling protocols significantly enhances the stability and efficiency of MnO₂-based electrodes for aqueous CO₂ capture. By implementing charge-limited cycling, the electrodes achieve over 98% capacity retention and double CO₂ throughput across 60 cycles, while also reducing energy requirements to near theoretical minimums, marking a significant advancement in electrochemical CO₂ separation technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] The Chemistry of Lithium Alkyl Carbonates in the Graphite SEI of Lithium-Ion Batteries** — score: 0.700 The article investigates the role of lithium alkyl carbonates in the solid electrolyte interphase (SEI) of lithium-ion batteries, highlighting their chemical interactions and implications for battery performance. Through analysis, it aims to improve understanding of the SEI's composition and stability, ultimately aiding in the development of more efficient battery technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Inhibiting Solvent Co-intercalation into Graphite Anode by Constructing an Anion-Rich Artificial Solid Electrolyte Interphase** — score: 0.700 This article discusses the development of an anion-rich artificial solid electrolyte interphase (ASEI) to inhibit solvent co-intercalation in graphite anodes. The study investigates how this innovative approach can enhance the stability and performance of lithium-ion batteries by preventing detrimental reactions that can compromise battery efficiency. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Theoretical Design of Mo₂TiC₂O₂ MXene as an Efficient Sulfur Immobilizer and Electrocatalyst for Lithium- and Sodium-Sulfur Batteries** — score: 0.700 The article discusses the theoretical design of Mo₂TiC₂O₂ MXene, highlighting its potential as an effective sulfur immobilizer and electrocatalyst for both lithium and sodium-sulfur batteries. The study emphasizes the material's promising properties that could enhance battery performance and energy storage efficiency. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Structure and Interfacial-Stable Binder Engineering Enables High-Rate Capability of NCM811 Cathodes at 4.6 V (Adv. Energy Mater. 21/2026)** — score: 0.600 Yong Qian and colleagues have designed a novel three-dimensional supermolecular binder that enhances the stability and performance of high-voltage NCM811 cathodes in lithium metal batteries. By using a crosslinked combination of biomass lignin and poly(acrylonitrile-co-acrylic acid), the binder not only scavenges harmful radicals and prevents metal ion dissolution but also maintains the integrity of the electrode structure, resulting in a pouch cell that achieves an energy density of 376.5 Wh kg⁻¹ and retains 81.7% capacity after 100 cycles at 4.6 V. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **In Situ Electronic Characterization of Electrochemically Tunable Metal Oxide Nanostructures for Energy Applications via Inverted Scanning Tunneling Spectroscopy** — score: 0.600 The article discusses innovative research on metal oxide nanostructures that can be electrically modified for energy applications, utilizing inverted scanning tunneling spectroscopy for in situ electronic characterization. The findings, aimed at advancing energy technologies, reveal potential improvements in the functionality and efficiency of these materials. Journal: *ScienceDirect Publication: Nano Energy*
- **Mechanisms of Alkali Ionic Transport in Amorphous Oxyhalides Solid State Conductors** — score: 0.600 This study employs large-scale machine learning molecular dynamics simulations to

explore the ionic conductivity of isovalent amorphous oxyhalides, revealing that their conductivity is nearly chemical composition-independent. The research identifies divalent anions as key obstacles to alkali ion diffusion and shows that despite significant ion correlations, the alkali transference numbers remain close to one, indicating efficient ionic transport across various compositions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Synergizing Phase Engineering and Site-selective Substitution Through Rational Design of High-Entropy Layered Oxide Cathode Enables Air-Stable and Long-life Sodium-ion Batteries** — score: 0.600 The article discusses a novel approach to enhancing the performance of sodium-ion batteries by integrating phase engineering and site-selective substitution in the design of high-entropy layered oxide cathodes. This strategy aims to address key challenges such as phase instability and degradation, ultimately enabling the development of air-stable, long-life batteries suitable for commercialization. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Coupled Solvation and Interfacial Regulation Enables High-Voltage and Low-Temperature Quasi-Solid Electrolytes for Ultra-High-Ni Cathodes** — score: 0.600 The article discusses advancements in quasi-solid electrolytes that demonstrate the ability to function at high voltages and low temperatures, which are critical for enhancing the performance of ultra-high-Ni cathodes in energy storage applications. The study highlights the importance of coupled solvation and interfacial regulation in achieving these enhancements, potentially leading to more efficient battery technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Achieving Kinetic Balance via Inorganic/Organic Dual-Phase SEI for Ultrastable Aqueous Zinc Metal Batteries** — score: 0.500 The article discusses the development of a dual-phase solid electrolyte interphase (SEI) that integrates both inorganic and organic materials to enhance the stability and performance of aqueous zinc metal batteries. This innovative approach aims to achieve kinetic balance, ultimately improving the longevity and efficiency of these energy storage systems. Journal: *ScienceDirect Publication: Nano Energy*
- **Boosting Pt Utilization via PtTe₂ Nanoshells for Direct Methanol Fuel Cells** — score: 0.400 The study presents a novel catalyst featuring PtTe₂ nanoshells on Te nanorods, which enhances platinum (Pt) utilization efficiency while reducing Pt loading to 5 wt.%. This catalyst demonstrates superior performance in direct methanol fuel cells, achieving a peak power density of 1.5 W mgPt⁻¹, significantly outperforming conventional Pt-based catalysts by multiple folds. Additionally, it minimizes CO poisoning effects, facilitating more efficient methanol oxidation through a predominantly direct pathway. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Nonplanar Tertiary-N Extended Nitrobenzene Enables Insoluble and Low-Energy-Barrier Organic Small Molecule Cathode for High-Performance Aqueous Batteries** — score: 0.400 The article discusses the development of a novel nonplanar tertiary-nitrogen extended nitrobenzene compound, which serves as an effective cathode material for aqueous zinc batteries. This organic small molecule demonstrates low solubility and a low energy barrier for ion transfer, enhancing battery performance by optimizing redox-active site exposure while minimizing adverse interactions with aqueous electrolytes. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Tailoring an iron-based metal-organic framework for a Zn-ORR/CER battery** — score: 0.400 This article discusses the development of a specialized iron-based metal-organic framework catalyst, Fe-Tp-10% G, designed for efficient chlorine production from hydrochloric acid while functioning as a cathode in a zinc-oxygen reduction/carbon dioxide evolution (Zn-ORR/CER) battery. The research emphasizes the catalyst's dual role in energy storage and conversion processes, highlighting its industrial relevance. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Boosting ionic conductivity of single-ion conductive polyelectrolyte elastomers via high-dielectric plasticizers** — score: 0.400 The article discusses the enhancement of ionic conductivity in polyelectrolyte elastomers through the incorporation of high-dielectric plasticizers. This innovation results in a twofold increase in ionic conductivity while preserving the material's elastic properties and preventing leakage, addressing a significant limitation of these materials. Journal: *Nature Materials*

- **[ASAP] Mitigating Succinonitrile–Li Molecular Crosstalk in In Situ Polymerization toward High-Voltage and Low-Temperature Solid-State Li Metal Batteries** — score: 0.400 The article discusses strategies to minimize the adverse effects of succinonitrile-lithium ion interactions during in situ polymerization processes, aiming to enhance the performance of solid-state lithium metal batteries for applications at high voltages and low temperatures. The research highlights the importance of addressing molecular crosstalk in order to improve battery efficiency and stability. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Unlocking High-Energy Metal Fluoride Cathodes through Modulated Interfacial Kinetics** — score: 0.400 The article explores innovative techniques for enhancing the performance of high-energy metal fluoride cathodes in batteries by manipulating interfacial kinetics. This research aims to unlock the full potential of these materials, improving energy storage capabilities and efficiency in battery technology. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Valence-Shell Electrons and Ionic Radius as Descriptors for Multisite Doping of RuO₂ for Durable Zn-Air Batteries** — score: 0.400 The article investigates the role of valence-shell electrons and ionic radius in optimizing multisite doping of RuO₂ to enhance the durability of Zn-air batteries. By analyzing these descriptors, the study aims to improve the electrochemical performance and longevity of battery materials. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Coupling Redox Compensation and Interfacial Stabilization in Low-Ni O₃-Type Sodium Layered Oxide Cathodes** — score: 0.400 The article discusses advancements in low-Ni O₃-type sodium layered oxide cathodes by integrating redox compensation with interfacial stabilization techniques. This approach aims to enhance the electrochemical performance and longevity of batteries, potentially addressing challenges associated with these materials in energy storage applications. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*
- **[ASAP] Cellulose-Derived Uniform N, P Codoped Carbon Aerogel as a High-Performance Modified Separator for Lithium–Sulfur Batteries** — score: 0.400 The article discusses the development of a cellulose-derived carbon aerogel that is codoped with nitrogen and phosphorus to enhance its performance as a separator in lithium-sulfur batteries. This innovative material demonstrates improved efficiency and durability, positioning it as a promising solution for next-generation energy storage technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Spin-State Engineering of Fe Single Atoms via Sulfur Incorporation into Carbon Matrix for High-Efficiency Oxygen Reduction and Zn–Air Batteries** — score: 0.400 The article discusses the enhancement of spin states in iron single atoms embedded within a sulfur-doped carbon matrix, resulting in improved efficiency for oxygen reduction reactions and zinc-air battery performance. The study emphasizes the potential of this approach in advancing energy storage technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Tailoring Closed-Pore Architectures in Biomass-Derived Hard Carbons for Superior Sodium Storage: From Structural Evolution Mechanisms to Pore Engineering** — score: 0.400 The article explores the design and optimization of closed-pore structures in hard carbons derived from biomass for enhanced sodium storage capabilities. It details the mechanisms of structural evolution and highlights strategies for pore engineering to improve the material’s performance in energy applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] A-Site Engineered High Entropy Perovskite Oxides for Rechargeable Zinc-Air Batteries** — score: 0.400 The article discusses the development of A-site engineered high entropy perovskite oxides designed to enhance the performance of rechargeable zinc-air batteries. By tailoring the composition of these materials, the study aims to improve battery efficiency and longevity, potentially advancing energy storage technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Electrochemically Induced Gradient Porous MXene/PEDOT:PSS Gel Electrodes**

- for High-Performance Energy Storage** — score: 0.400 The article discusses the development of electrochemically induced gradient porous gel electrodes made from MXene and PEDOT:PSS, which significantly enhance energy storage performance. This innovative approach addresses challenges in energy storage systems by optimizing electrode structure and composition, leading to improved efficiency and effectiveness in energy applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Interface regulation empowering V2O5-based cathode for high-rate aqueous zinc-ion batteries with ultra-long cycle life** — score: 0.400 The article discusses the development of a V2O5-based cathode designed for high-rate aqueous zinc-ion batteries, emphasizing the role of interface regulation in enhancing battery performance. The findings indicate that this approach significantly extends the cycle life of the batteries, presenting a promising advancement for energy storage technologies. Journal: *ScienceDirect Publication: Nano Energy*
 - **Customized solvation of gel polymer electrolyte with fluoroacetonitrile solvent for wide temperature solid-state lithium metal battery** — score: 0.300 The article discusses a novel gel polymer electrolyte that utilizes fluoroacetonitrile as a solvent, tailored for solid-state lithium metal batteries. This approach aims to enhance the temperature range and performance of the batteries, potentially leading to advancements in energy storage technology. Journal: *ScienceDirect Publication: Nano Energy*
 - **[ASAP] Laser-Induced Three-Dimensional Zn_{1-x}Cu_xO Solid-Solution Protective Layer for High-Stability Aqueous Zinc-Ion Batteries** — score: 0.300 This article discusses the development of a novel laser-induced protective layer made from a Zn_{1-x}Cu_xO solid solution, aimed at enhancing the stability of aqueous zinc-ion batteries. The innovative approach addresses common stability issues in these batteries, promising improved performance and longevity. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
 - **[ASAP] Boosting Ionic Conductivity in Zirconium-Based Oxychloride Solid Electrolytes via Br Doping for All-Solid-State Batteries** — score: 0.300 The article discusses a method to enhance ionic conductivity in zirconium-based oxychloride solid electrolytes for all-solid-state batteries through bromine doping. This improvement aims to optimize battery performance and efficiency, addressing limitations in current solid electrolyte materials. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
 - **[ASAP] Active Interfacial Solvation Architecture for Highly Reversible Zinc Metal Anodes** — score: 0.300 The article discusses the development of a novel interfacial solvation architecture that enhances the reversibility and stability of zinc metal anodes in batteries. This innovative approach addresses challenges related to zinc deposition and dissolution, thereby improving the performance of zinc-based energy storage systems. Journal: *ACS Energy Letters: Latest Articles (ACS Publications)*